

Data Science

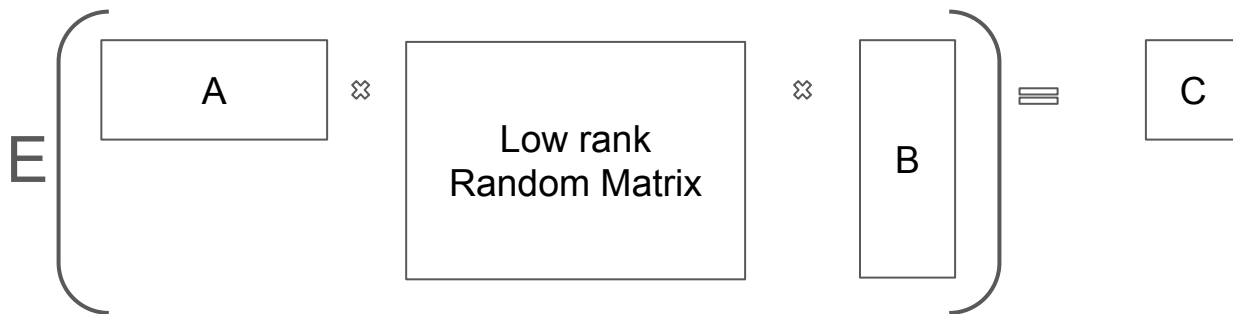
CSE558

Recap

- Low Rank Approximation

$$\|X - \mathbf{x}\mathbf{v}_k\mathbf{v}_k^T\|_F^2 = \sum_{i=k+1}^d \lambda_i(\mathbf{X}^T\mathbf{X})$$

- Fast Matrix Multiplication



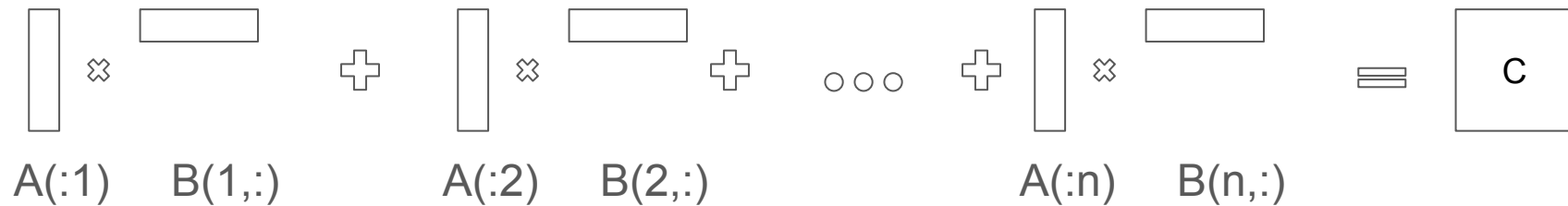
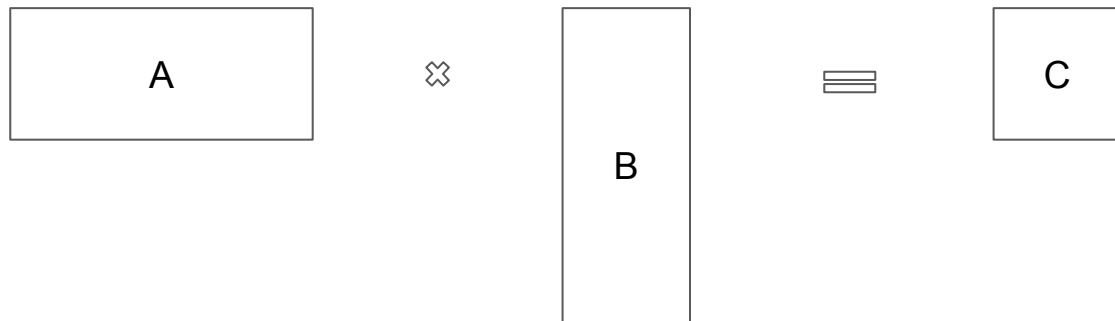
Questions

1. Consider a corpus of 1M documents each having 100k words from a vocabulary of size 200k words. Design a preprocessing step to ensure that the size of the data is small.
2. If the task is topic modeling (uses pairwise distance between documents), then given a 3 sided unbiased die, design a family of JL matrix that can be used for topic modeling.

Overview

- Fast Matrix Multiplication
- Matrix Completion

Sampling



$$AB = \sum_{k=1}^n A(:,k)B(k,:)$$

Sampling

Consider the following random matrix

$$X = \left\{ \frac{1}{p_k} A(:, k) B(k, :), \quad \text{with probability } p_k \right.$$

$$E[X] = \sum_{i=1}^n \frac{1}{p_i} A(:, k) B(k, :) \cdot p_i = \sum_{i=1}^n A(:, k) B(k, :) = AB$$

Proper scaling, gives unbiased results!

Works for any distribution?

Sampling

Unbiased is not enough!

Analyze the guarantee of the random experiment.

$$E(\|AB - X\|_F^2) = \text{Var}(X)$$

$$\text{Var}(X) = \sum_{i=1}^m \sum_{j=1}^p \text{Var}(x_{ij}) = \sum_{ij} E(x_{ij}^2) - E(x_{ij})^2 = \left(\sum_{ij} \sum_k p_k \frac{1}{p_k^2} a_{ik}^2 b_{kj}^2 \right) - \|AB\|_F^2$$

We can only control the first term!

$$\sum_{ij} \sum_k p_k \frac{1}{p_k^2} a_{ik}^2 b_{kj}^2 = \sum_k \frac{1}{p_k} \left(\sum_i a_{ik}^2 \right) \left(\sum_j b_{kj}^2 \right) = \sum_k \frac{1}{p_k} |A(:, k)|^2 |B(k, :)|^2$$

Sampling

Unbiased is not enough!

Analyze the guarantee of the random experiment.

$$E(\|AB - X\|_F^2) = \text{Var}(X) \quad \text{Var}(X) = \left(\sum_{ij} \sum_k p_k \frac{1}{p_k^2} a_{ik}^2 b_{kj}^2 \right) - \|AB\|_F^2$$

$$\sum_{ij} \sum_k p_k \frac{1}{p_k^2} a_{ik}^2 b_{kj}^2 = \sum_k \frac{1}{p_k} \left(\sum_i a_{ik}^2 \right) \left(\sum_j b_{kj}^2 \right) = \sum_k \frac{1}{p_k} |A(:, k)|^2 |B(k, :)|^2$$

$$\text{Let, } p_k = \frac{|A(:, k)|^2}{\|A\|_F^2}$$

$$\sum_{ij} \sum_k p_k \frac{1}{p_k^2} a_{ik}^2 b_{kj}^2 = \|A\|_F^2 \sum_k |B(k, :)|^2 = \|A\|_F^2 \|B\|_F^2$$

Sampling

$$E(\|AB - X\|_F^2) = \text{Var}(X) \leq \|A\|_F^2 \|B\|_F^2$$

Can we do better?

Take S independent random matrices, $X_1, X_2, \dots, X_s$ and return $\frac{1}{s} \sum_{i=1}^s X_i$

$$E \left[\left(AB - \frac{1}{s} \sum_{i=1}^s X_i \right)^2 \right] \leq \frac{\|A\|_F^2 \|B\|_F^2}{s}$$

When is this a good approximation and when not?

Sampling

Sampling not helpful

Let $A = B = I_n$.

Then, $\|AB\|_F^2 = n$ and $\|A\|_F^2 \|B\|_F^2 = n^2$

$$E \left[\left(AB - \frac{1}{s} \sum_{i=1}^s X_i \right)^2 \right] \leq \frac{\|A\|_F^2 \|B\|_F^2}{s}$$

To get a useful bound, we need $s > n$.